

Statespace (S)

s_1 : Exhausted

s_2 : Prepared

s_3 : hanging out / Socially happy
done

Action Space (A)

a_1 : Study

a_2 : Sleep

a_3 : Hang out

Transition Probabilities

- From (s_1)

→ a_2 leads to s_2 with $P=1.0$

- From (s_2)

→ a_1 → (s_2) $P=1.0$

- From (s_3)

→ (a_1) → (s_1) with $P=1.0$

Reward Function (R)

- $R(s_1, a_2) = +5$ (sleeping when exhausted is good)
- $R(s_2, a_1) = +10$ (studying while alert is great long term)
- $R(s_3, a_1) = -2$ (very tough to study after hang out)

Eqn.

$$V_{k+1}(s) = \max_a \left[R(s, a) + \gamma \sum P(s' | s, a) V_k(s') \right]$$

$$\gamma = 0.9, \quad V_0(s_1) = 0, \quad V_0(s_2) = 0, \quad V_0(s_3) = 0$$

$k=1$ Iteration 1

→ For s_1 : $V_1(s_1) = R(s_1, a_1) + 0 = 5$

→ For s_2 : $V_1(s_2) = R(s_2, a_1) + 0 = 10$

→ For s_3 : $V_1(s_3) = R(s_3, a_1) + 0 = -2$

Now,
 $V_1(s_1) = 5$
 $V_1(s_2) = 10$
 $V_1(s_3) = -2$

$k=2$

→ For s_1 :

$$V_2(s_1) = R(s_1, a_2) + \gamma \cdot V_1(s_2) = 5 + 0.9(10) = 14$$

→ For s_2 :

$$V_2(s_2) = R(s_2, a_1) + \gamma \cdot V_1(s_2) = 10 + 0.9(10) = 19$$

→ For s_3 :

$$V_2(s_3) = R(s_3, a_1) + \gamma \cdot V_1(s_3) = -2 + 0.9(5) = 2.5$$

Final Values

$$V_2(s_1) = 14$$

$$V_2(s_2) = 19$$

$$V_2(s_3) = 2.5$$